

Access Reader — Scope Document v3

AI Forum Hackathon 2026 · NZSL pivot

Problem Statement

New Zealand Sign Language has been an official language of Aotearoa since 2006. Section 9(1)(c) of the NZSL Act says government services and information should be made accessible to the Deaf community, including through NZSL. Twenty years on, almost none of the documents that actually govern people's lives — rates notices, benefit letters, tenancy agreements, medical forms — exist in NZSL. They never will, because you cannot book a human interpreter for every letter that lands in every letterbox.

Access Reader turns a written document into NZSL. Upload or scan it, and it comes back as signed content with plain-language subtitles.

Who We're Solving For

Deaf NZSL users in Aotearoa. ~23,000 people have some knowledge of NZSL; an estimated 3,000–4,000 are Deaf people for whom it is their primary language.

The critical framing: NZSL is a distinct language, not signed English. For a Deaf person whose first language is NZSL, an English document is a **second-language document** — not just small print. This is a translation problem, not a formatting problem. Say this in the pitch; it reframes the whole product.

What We Are Explicitly Not Claiming

The World Federation of the Deaf and WASLI have cautioned against signing avatars as a **replacement for human interpreters**, because word-to-sign translation isn't possible without context and cultural norms, and machine translation hasn't matched human ability. The WFD position is also that Deaf people should lead development of AI tools using their national sign languages — and we have no Deaf team members.

We accept both points and design around them. Our positioning:

"We're not replacing NZSL interpreters — the Deaf community's own peak body has warned against exactly that, and they're right. We're closing the gap for the millions of everyday documents that will never get an interpreter at all. The next step isn't more engineering, it's building this with Deaf Aotearoa, not for them."

Have this answer ready verbatim. A judge who knows this space **will** ask.

Why the Avatar Has a Face (technical decision, pitch it)

We considered a hands-only avatar and rejected it. In NZSL, facial expression is grammar, not emotion — the Ministry of Education's own NZSL resource calls it "facial grammar", the equivalent of vocal intonation. Non-manual features carry negation, question marking, topic marking and grammatical mood. Research indicates non-manual markers carry up to 30% of sign meaning; in one test, distinguishing two similar signs from hand landmarks alone gave 63% accuracy, and adding facial features cut confusion from 37% to 11%.

In plain terms: without a face, "you do not need to do this" and "do you need to do this?" can render identically. On a benefit letter, that is a harm, not a bug.

This is a strong "Design and Innovation" beat — it shows we understood the linguistics rather than just rendering hands.

Technical Architecture

Document / page



OCR layer → detect and extract content



Cache / database → extracted data cloned and stored, OCR runs once



AI pass 1: SIMPLIFY → plain-language English, short sentences



AI pass 2: GLOSS → NZSL gloss sequence + non-manual markers

(brow raise = y/n question, brow furrow = wh-,

headshake = negation, topic marking, mouth patterns)



Render layer → avatar animates gloss with face + hands



Output → signing video + running subtitles + simplified text

Pitch the cache as a deliberate engineering decision: OCR runs once, then every replay, re-watch and section jump reads from the database. A Deaf user re-watching a clause five times shouldn't wait for five OCR passes.

Pitch the gloss layer as the actual innovation: the AI doesn't translate English words to signs one-for-one — it produces NZSL gloss *with non-manual annotation*, because that's where the grammar lives. That annotation is what drives the face.

Feature Scope

Building today (must demo)

- OCR scan/extract → cache to database
- AI simplify pass (plain-language English)
- AI gloss pass (NZSL gloss + non-manual markers)
- Avatar playback with **face and hands visible**
- Running subtitles beneath the avatar
- Section-by-section navigation (see UX note below)

Visible in UI, marked "Coming soon" (do not build)

- **What's missing?** — detects unfilled fields and absent required information
- **Live camera detection** — sign from the camera rather than a static upload

Roadmap only (say, don't build)

- Deaf-led validation and co-design with Deaf Aotearoa
- Regional NZSL variation and dialect handling
- Expanded document types

The UX Insight That Should Drive the Interface

Sign video is linear. You cannot skim it. This is the exact problem blind screen-reader users have with audio — one-dimensional narration, no visual scanning. If we render a four-page tenancy agreement as one unbroken signing video, we've rebuilt the barrier in a new medium.

So: chunk by section, with jump navigation. The user sees a list of sections — "What this is", "What you need to do", "By when", "What happens if you don't" — and jumps straight to the one they need. Replay any section without restarting.

This is the single design decision most likely to win Design and Innovation points, because it's the one that comes from understanding the user rather than the technology.

Feasibility Reality Check (decide by 2pm Scope checkpoint)

Full text-to-NZSL avatar generation is a research problem, not a weekend problem. Pick a lane now:

Approach	Effort	Risk
A. Pre-rendered demo path — one document, signing pre-generated, live pipeline real up to the gloss stage	Low	Must disclose it's a scripted demo
B. Gloss → stitched clip playback — AI generates gloss live, front end plays back a small library of pre-recorded signs	Medium	Library size limits vocabulary; stitching looks choppy
C. Live 3D avatar with blendshape face — rig driven by gloss + non-manual markers	High	Very likely to eat the whole 48 hours

Recommendation: B, with A as the fallback. B keeps the AI visibly load-bearing (the gloss generation is real and live), which is what "Technology" is scored on, while keeping the render layer achievable. Cap the sign library to whatever one realistic demo document needs.

Disclosure rule: whatever you pick, say plainly in the pitch what is live and what is pre-rendered. Judges have seen a hundred hackathon demos and respect the team that says it. They do not respect the team that gets caught.

Known Risks to Address Before You're Asked

1. **Translation accuracy.** AI gloss will get things wrong on legal and medical language.
Mitigation: the original English text stays visible alongside the signing at all times. We never replace the source, we sit beside it.

2. **No Deaf input.** We built this in 48 hours with no Deaf team members. **Mitigation:** say so first, and make Deaf-led co-design the explicit next step rather than a footnote.
3. **Regional variation.** NZSL has dialect and vocabulary variation. **Mitigation:** acknowledge it; scope v1 to a single variety.

Design Principles

1. **The prototype must itself be accessible.** No audio-only cues anywhere — our users can't hear them.
2. **Never hide the original.** Trust is the product.
3. **Frame the avatar properly.** Head to waist, hands free to move above the head. Never crop the face or the signing space.
4. **Plain contrasting background** behind the avatar, following interpreter convention. No busy patterns competing with the hands.
5. **No disability iconography.** This is a language tool, not a medical device.

Ownership

Area	Owner
OCR + database/cache	
AI simplify + gloss pipeline	
Avatar render + playback	
UI / prototype	
Pitch	

Checkpoints

- **2:00pm — Scope:** feasibility lane locked (A, B or C). Demo document chosen and frozen.
- **5:30pm — It's Alive:** OCR → cache → simplify → gloss working end to end, even if the avatar is still rough
- **7:00pm — Pitch skeleton:** 5-minute structure mapped, WFD answer rehearsed, practised out loud with a timer